Light

GCSE Computer Science — Boolean Logic (Set 3)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What output does a NOT gate produce when the input is 1?

- A. 0
- B. 1
- C. 2
- D. The input value unchanged

Q2 (1 mark)

How many inputs does a standard AND gate have?

- A. 1 input

- **B.** 2 inputs
- **C.** 3 inputs
- **D.** Any number of inputs

Q3 (1 mark)

What is the ONLY input combination that produces output 0 from an OR gate?

- **A.** Both inputs are 0
- **B.** Both inputs are 1
- **C.** First input is 0, second input is 1
- **D.** First input is 1, second input is 0

Q4 (1 mark)

What shape is the main body of a NOT gate symbol?

- **A.** A rectangle
- **B.** A triangle
- **C.** A semicircle
- **D.** A diamond

Q5 (1 mark)

How many rows would a truth table have for a logic circuit with 4 inputs?

- ○ **A.** 4 rows
- ○ **B.** 8 rows
- ○ **C.** 16 rows
- ○ **D.** 32 rows

Q6 (1 mark)

What is the purpose of a logic gate in a digital circuit?

- ○ **A.** To perform a Boolean operation on input signals
- ○ **B.** To store data permanently
- ○ **C.** To convert analogue signals to digital
- ○ **D.** To amplify weak electrical signals

Q7 (1 mark)

A vending machine dispenses a drink only when money is inserted (M=1) AND a valid selection is made (S=1). Which gate models this?

- ○ **A.** NOT gate
- ○ **B.** OR gate
- ○ **C.** AND gate

- ○ **D.** No gate needed

Q8 (1 mark)

A house has two doorbell buttons - front (F) and back (B). The bell rings if the front button is pressed OR the back button is pressed. Which gate represents this?

- ○ **A.** AND gate
- ○ **B.** OR gate
- ○ **C.** NOT gate
- ○ **D.** Two separate gates are needed

Q9 (1 mark)

A fridge light turns ON (output=1) when the door is NOT closed (door sensor=0). Which gate produces this?

- ○ **A.** AND gate
- ○ **B.** OR gate
- ○ **C.** NOT gate
- ○ **D.** The light connects directly without a gate

Q10 (1 mark)

In a logic circuit diagram, what connects the output of one gate to the input of another?

- **A.** Arrows only
- **B.** Nothing - gates must be adjacent
- **C.** Dotted paths
- **D.** Lines (wires)

Q11 (1 mark)

Can the output of a Boolean logic gate be a value like 0.5?

- **A.** Yes, if the inputs are mixed
- **B.** Yes, in some special gates
- **C.** No, outputs can only be 0 or 1
- **D.** Only in analogue circuits

Q12 (1 mark)

What is another common name for a NOT gate?

- **A.** Amplifier
- **B.** Inverter
- **C.** Buffer
- **D.** Comparator

Q13 (2 marks)

Evaluate: (1 AND 0) OR (0 OR 1)

- ○ **A.** 0
- ○ **B.** 2
- ○ **C.** Cannot be determined
- ○ **D.** 1

Q14 (2 marks)

What is the output of NOT (NOT 0)?

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** -1
- ○ **D.** Undefined

Q15 (2 marks)

Evaluate: (0 AND 0) OR (1 AND 1)

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** Both 0 and 1

- **D.** Neither 0 nor 1

Q16 (2 marks)

Evaluate: (1 OR 0) AND (0 OR 0)

- **A.** 0
- **B.** 1
- **C.** Depends on evaluation order
- **D.** Error

Q17 (2 marks)

In a truth table for A AND B, the row shows A=1, B=1. What is the output?

- **A.** 0
- **B.** Cannot determine from this row alone
- **C.** Either 0 or 1
- **D.** 1

Q18 (2 marks)

In a truth table for A OR B, the row shows A=0, B=1. What is the output?

- **A.** 0

- **B.** Same as B
- **C.** Same as A
- **D.** 1

Q19 (2 marks)

A logic circuit is B AND (NOT A). If A=0 and B=1, what is the output?

- **A.** 0
- **B.** Opposite of B
- **C.** Same as A
- **D.** 1

Q20 (2 marks)

Which Boolean expression outputs 1 ONLY when A=1 AND B=0?

- **A.** A OR B
- **B.** A AND B
- **C.** A AND (NOT B)
- **D.** (NOT A) AND B

Q21 (2 marks)

Evaluate: (NOT A) AND (NOT B), where A=1, B=1

- **A.** 0
- **B.** 1
- **C.** Cannot calculate with two NOTs
- **D.** Depends on the order of operations

Q22 (2 marks)

Which expression is equivalent to NOT A? (Test by checking: when A=0 it outputs 1, when A=1 it outputs 0)

- **A.** A AND A
- **B.** A OR A
- **C.** A AND (NOT A)
- **D.** None of the above - only NOT A produces this pattern

Q23 (2 marks)

An ATM dispenses cash when: the card is valid (C=1) AND the PIN is correct (P=1) AND sufficient funds exist (F=1). Which expression models this?

- **A.** C OR P OR F
- **B.** C AND P AND F
- **C.** (C AND P) OR F
- **D.** C OR (P AND F)

Q24 (3 marks)

A logic circuit has two inputs P and Q. The complete truth table is:

P	Q	Output
0	0	0
0	1	1
1	0	1
1	1	1

Which Boolean expression produces this truth table?

- ○ **A.** P AND Q
- ○ **B.** P OR Q
- ○ **C.** NOT P
- ○ **D.** NOT (P AND Q)

Q25 (3 marks)

A logic circuit is: A AND (NOT B OR C).
For which input combination does the output equal 1?

- ○ **A.** A=0, B=0, C=0
- ○ **B.** A=1, B=1, C=0
- ○ **C.** A=1, B=0, C=0
- ○ **D.** A=0, B=1, C=1

Q26 (3 marks)

A school PA system broadcasts announcements when:

- The principal authorises (P=1), AND
- Either the main speaker works (M=1) OR the backup speaker works (B=1)

Which expression correctly models this system?

- ○ **A.** P OR M OR B
- ○ **B.** P AND M AND B
- ○ **C.** P AND (M OR B)
- ○ **D.** (P AND M) OR B

Q27 (3 marks)

Complete the truth table for: A AND (NOT B)

A	B	Output
0	0	?
0	1	?
1	0	?
1	1	?

Which row pattern shows the correct outputs (top to bottom)?

- ○ **A.** 0, 0, 0, 1
- ○ **B.** 0, 0, 1, 0
- ○ **C.** 1, 0, 1, 0
- ○ **D.** 0, 1, 0, 1

Q28 (3 marks)

A student designs a circuit to control a motor. The motor should run when BOTH the power switch is ON (P=1) AND the safety guard is closed (G=1).
The student incorrectly writes: P OR G
What is wrong, and what dangerous situation could occur?

- ○ **A.** The motor won't run even when both conditions are met
- ○ **B.** The motor runs when ONLY the power is on, even with the safety guard open
- ○ **C.** The motor runs backwards
- ○ **D.** The circuit has no errors

Q29 (3 marks)

A factory robot arm operates under these conditions:
Operate = (Sensor clear (S=1) AND Program running (R=1)) OR Emergency stop NOT pressed (E=0)
Given: S=1, R=0, E=1 (emergency stop IS pressed)
Does the robot operate?

- ○ **A.** Yes, because the sensor is clear
- ○ **B.** Yes, because emergency stop being pressed activates the robot
- ○ **C.** No, because the program isn't running and emergency stop is pressed
- ○ **D.** Cannot determine without additional information

Q30 (3 marks)

A logic circuit uses the expression: (NOT A OR B) AND (NOT A OR NOT B).

Analyse this expression. What simpler equivalent expression produces the same outputs?

- ○ **A.** A AND B
- ○ **B.** A OR B
- ○ **C.** NOT A
- ○ **D.** B

Total: **55** marks • Time: **30 min**

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